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ROAD DAMAGE DETECTION AND CLASSIFICATION USING YOLOv5

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***Abstract*— Road infrastructure is a critical public asset that must be inspected and monitored regularly since it contributes to economic progress. However, road surface deteriorates over time from various factors. Detecting road damage quickly and precisely allows road-maintenance agencies to conduct timely repairs, maintain optimal road conditions,and improve transportation safety. For proper road maintenance, detection should be automated to eliminate time-consuming and inefficient manual inspection.The scope of automatic pavement defect assessments has significantly improved over the years. In this paper, we propose road damage detection and classification system that uses YOLOv5 to automatically detect damages from the road images. The model considers seven categories comprising mainly cracks , namely D00, D10, D20, D40, D43, D44 and D50. Evaluation results show that the overall accuracy of the road damage detection and classification model is 92%. We illustrate how YOLOv5, one of the most recent state-of-the-art object detection and classification algorithms, can be utilized to perform this task in a fast manner with effective results.**

***Index Terms*— Damage Detection, YOLOv5, Classification, Crack, RDD**

I. INTRODUCTION

The Road infrastructure of a country is considered to be of utmost importance in terms of its social and economic significance. Therefore, governments sets aside large amounts of money each year to repair and maintain road networks. Roads, like any other sort of infrastructure,are susceptible to damage from a variety of causes such as weather, traffic accidents, ageing and moisture infiltration. As a result, various forms of distress such as cracks, potholes, and corrugation begin to appear on the road's surface. Governments require road inspection systems to assess road surface conditions on a regular basis.

Road damage can be evaluated using three methods: manual, semi-automated, or fully automated [1]. In manual approach, assessors perform a visual inspection of the road surface and determine the condition of pavements and give off a rating that is subject to vary depending on every other assessor. These manual inspection also require domain expertise but is time-consuming and unsafe. In semi-automated pavement condition assessments, road images are collected automatically from a vehicle, but distress identification is deferred to an offline process performed at the control station [1].

To overcome these problems, optimal and cost-effective steps could be applied that automate the detection and classification process. Automated evaluation techniques utilizing commercial grade cameras are one of them and have been used because of their low cost and easy accessibility [5].

With the technology evolving much faster than ever,we have several object detection methods based on Convolution neural network. One-stage detectors and Two-stage detectors comes under the category of object detection algorithm. Two-stage detectors such as R-CNN ,Fast R-CNN ,Faster R-CNN are composed of two steps: object region proposal followed by region classification. The one-stage detectors propose bounding boxes automatically from input images without the need for preliminary step,that makes time efficient and useful for real-time devices [11]. YOLO or You Only Look Once is a popular one-stage detector . Several versions of YOLO have been proposed,the most recent of which is the YOLOv5 [7] ,which has fast detection speed and high accuracy .

The main aim of the paper is to develop a YOLOv5 based road damage detection model that can efficiently detect and classify seven road damage types with high accuracy. YOLOv5 is latest version of YOLO model ,it has fast detection speed and can effectively work in real-time. The YOLOv5 model has also the advantage of adaptive generation of anchor boxes that helps to accurately fix the bounding boxes on the road damage.

The rest of the paper is organized as follows. In Section II, discussed about the related works. Section III presents the proposed methodology.Section IV explains the dataset. The experimental settings are explained in Section V. Finally, Section VI concludes the paper.

II. RELATED WORKS

Several methods were proposed for road damage detection and classification. Most of them focus on implementing two-stage detectors which has proven to be slow on real time detection as compared to one- stage detectors such as YOLO.

Deeksha Arya et al.[1] proposed a solution that utilizes SSD MobileNet model pretrained on MSCOCO dataset to train the detection model. The model can only classifies four road damage categories. Therefore,the generalized model can be utilised for monitoring road conditions in multiple countries. The proposed dataset includes road images taken from countries such as Japan,India and Czech Republic to make the data diverse and train robust algorithms.

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Yuchuan Du et al.[3] proposed YOLOv3 for pavement distress detection and classification. The dataset is composed of 45,788 images. The images were taken using industrial cameras mounted on the vehicle. The model uses transfer learning to avoid overfitting. The detection accuracy reached 73.64% for 7 road damage types. Additionally, they divided the dataset based on various image illuminations to identify how the model can distinguish different types of PD in different light conditions. YOLOv3 performed better when compared with the faster RCNN and SSD. It has a faster processing speed and similar accuracy to the Faster RCNN.

Ronald Roberts et al. [8] aimed at the automation of the pavement detection system to analyse distress that is present on urban roads. The Faster R-CNN model and the SSD model with InceptionV2 and MobileNetV2 CNNs were the two main networks used in this study. The proposed model can conduct regular monitoring of road damage detection at a low cost. The model was developed based on 7000 local images taken in Italy. Data augmentation was also applied to increase the dataset. The images were taken using a low-cost smartphone. The model can identify road damage types as well as their associated severities (Level 1 and Level 2). At level 1, remedial action is not compulsory, and at level 2, action is a necessity. The input images were of pixels of 300 x 300. Out of the three models evaluated, the Faster RCNN performed better. Additionally, they have implemented a pipeline showing how the model can be deployed in real-world scenarios. The road images would be stored in the database, along with the type of damage and severity level. Not only that, the location, length, and area of the road covered can also be stored. This further helps the road agency to note the area that requires quick maintenance.

Minh-Tu Cao et al. [10] compared the performance of eight road damage detection models. Faster RCNN and SSD object detection models were utilised. For feature extraction, they employed MobileNet, Inception v2, Resnet-50 and-100, and Inception Resnet. Performance was assessed on a single data source (RDD 2018) as well as a combination of multiple data sources. mAP value dropped significantly when they increased IoU from 0.5 to 0.75, therefore concluded that an IoU value of 0.5 is appropriate for road damage detection. The results showed Faster RCNN based models were better than SSD based models. But as Faster RCNN comes under the category of two stage detector, they were significantly slower than SSD based models. Overall performance results on the RDD 2018 shows that in terms of detection accuracy, Faster RCNN Inception V2 is preferred whereas SSD Inception V2 is recommended in terms of processing time.

Kortmann et al. [12] utilised the YOLOv5 framework to develop the best possible DL model for solving the detection problem of road damage for autonomous vehicles. They have only considered four types of road damage. So, the results showed that construction joints are likely to be classified as cracks. Due to this limitation, the other potential tool could be to apply more classes to the dataset.

III. PROPOSED METHODOLOGY

In this section, we first explain the principle of YOLO algorithm. Then the we introduce in detail the implementation of YOLOv5 algorithm.

A. Principle of YOLO algorithm

YOLO is a single-stage deep learning system that detects objects using a convolutional neural network. The basic principle of the YOLO is to view the complete image as input to the system and directly predict the bounding box and the category which it belong to. YOLO consider image detection as a regression problem.

The latest and lightweight version is the YOLOv5. It is written in Python programming language and uses Pytorch framework.

The YOLOv5 road damage detection model is portrayed in Figure 1. The road damage image is divided into a given number of grid cells, after which bounding boxes and a class probability map are generated. To remove excess bounding boxes,we need to use NMS. Following that, the final image with damage detection and classification is displayed.The detailed steps are described in the following sections.

1) Dividing the image: YOLO divides the input image into an S×S grid. Each grid cell predicts only one object. Each grid cell predicts a fixed number of bounding boxes.For each grid cell,

- it predicts B bounding boxes with one box confidence score for each,
- Only one object is detected irrespective of the number of boxes B,
- C conditional class probabilities for every object class

The objectness of the box and the accuracy of the boundary box are expressed in the confidence score. The values of x, y, w, and h should be between 0 and 1. Each cell have C conditional class probabilities which tells us that the chance of identified object belonging to a specific class .

2) Generation of Bounding Boxes: Each cell can predict a fixed B number of bounding boxes. The centres of the bounding box, width, and height are represented by x, y, w, and h, respectively. Each bounding box would also have confidence score. After the prediction,YOLO employs IOU (Intersection Over Union) to select the right bounding box of an object for the grid cell .

3) Generation of Class Probability Map: If the grid cell contains an object, it predicts conditional class probabilities. During test time, multiplying conditional class probabilities and individual box confidence score yields confidence score related to the specific class.

4) Applying Non Maximum Suppression: When more than one bounding box detects the object as a positive class detection,the same object gets detected multiple times. Non-maximum suppression (NMS) helps avoid this situation by filtering out excess boxes. YOLOv5 uses NMS to filter out bounding box with lower threshold value. The whole filtering process depends on single threshold value. As a result, choosing a threshold value is critical to the model's success.

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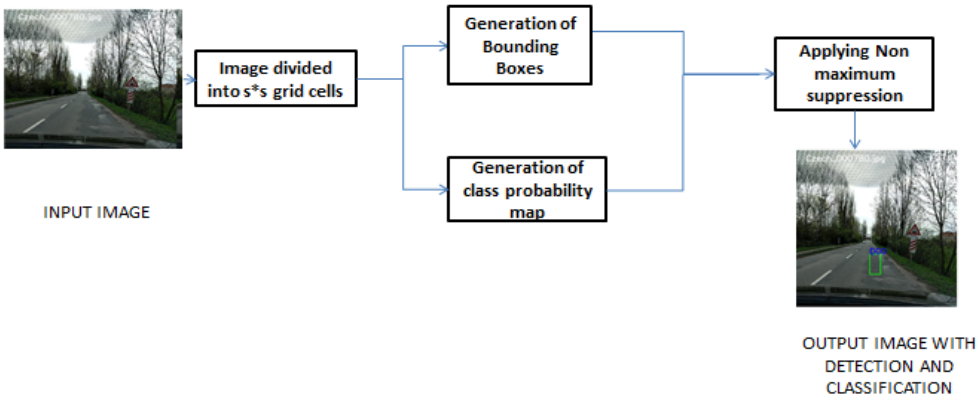


Fig. 1: Road Damage Detection model of YOLOv5

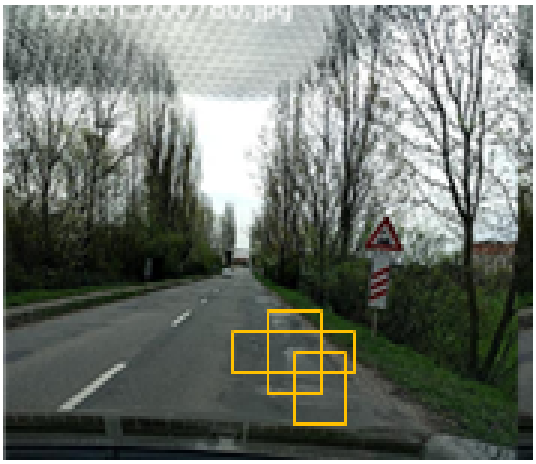


Fig. 2: Bounding Boxes

After applying the NMS,it filters the excess bounding box and there will be single bounding box for a object as shown in figure 3 .

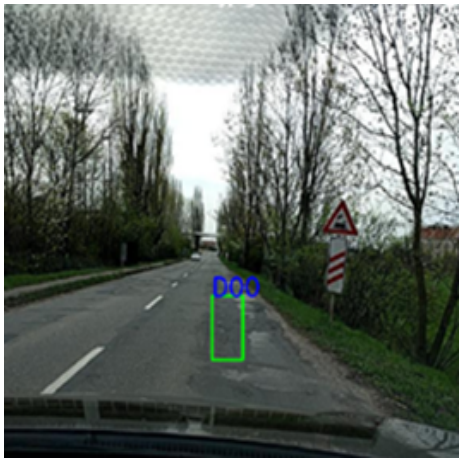


Fig. 3: Road damage detection and classification of D00

B. Implementation of YOLOv5 Algorithm

The proposed system aims to detect and classify road damages. The prediction and classification are both done using YOLOv5. Figure 4 depicts the flowchart for training the YOLOv5 model.

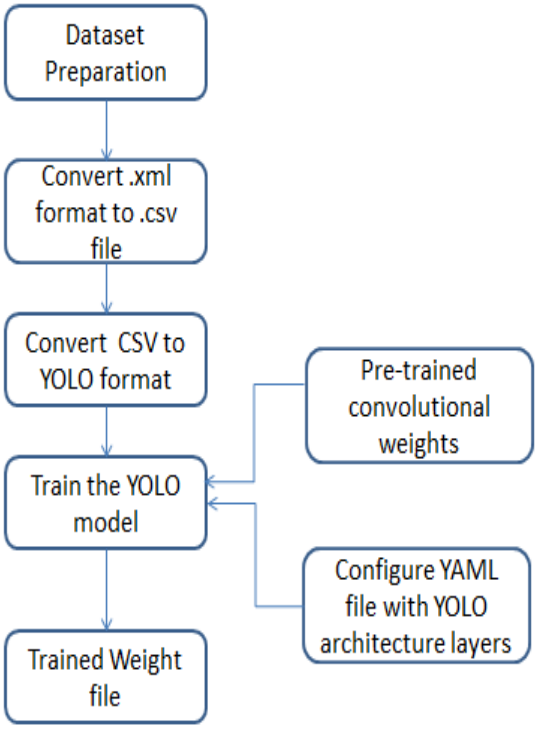


Fig. 4: Flowchart for training the YOLOv5 model

1) Dataset Preparation: The road images from Japan were used to train the model for the proposed concept. D00, D10, D20, D40, D43, D44, and D50 are the seven damage types that may be classified using this dataset.

2) Convert XML format to CSV format: The training dataset contains Japan road images as well as its corresponding annotations which is in pascal VOC format. The XML

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file has details regarding the object class found in the image along with its bounding box coordinates. Each xml file have to be converted to csv file format.

3) Convert CSV to YOLO format: YOLOv5 expects annotations for each image in the form of a .txt file, with each line of the text file defining a bounding box. Each line represents one of the road damage class and bounding box coordinates.

4) Train the YOLO model: To train the model, we need to configure a YAML file which includes the number of classes as well as names of the classes. The further details are explained in section V.

IV. DATASET

The Road Damage Dataset 2020 was collected from three countries: Czech Republic, India, and Japan. For the proposed work,we have trained the model only using Japan road images. Our dataset has 2607 images. As the YOLOv5 can work effectively on medium-sized dataset [7]. The dataset containing Japan road images can classify seven types of road damages. One or more classes of damages are labeled on each image. Table I shows the class name and road damage type that are based on the classification proposed by Maeda et al. [6]. Some of the training images don't have any road damage class; thus they are considered as negative images.

TABLE I: Road Damage Types

Class Name	Damage Type
D00	Perpendicular crack
D10	Traverse / Parallel Crack
D20	Alligator crack
D40	Potholes
D43	Crosswalk Blur
D44	White Line Blur
D50	Utility hole

V. EXPERIMENTS AND RESULTS

This section illustrates the results obtained for the proposed model. The evaluation metrics used shows the performance of the model.

A. Training

The model was trained in the Google Colab GPU runtime environment platform using Tensorflow. The operating system is windows 10. The model is used to detect and classify seven type of road damages. YOLOv5 object detector is trained on Japan road images. The dataset is divided into two, such as Training set and Validation set. The 80% of data is used for training and 20% of data for validation. The training dataset contains 2086 images and validation data includes 521 images. The original image size is 600 x 600,which is compressed to 416 x 416 pixels to make training process faster. To attain the best prediction outcome, YOLOv5l (YOLOv5 large) model is used and trained for 75 epochs using YOLOv5l pre-trained weight. Then,again trained with custom pre-trained best weights that resulted in 92% accuracy.

B. Evaluation

Evaluation of the object detection model can be based on various parameters. Here, to assess the quality of the detection model,the evaluation is based on the Precison, Recall and average accuracy value (mAP). Among these, mAP is often used to measure the performance of the object detection model. The greater the mAP score, the better the average detection accuracy and performance will be. Table II shows the mAP scores for specific damage types as well as overall accuracy.

TABLE II: mAP scores for specific damage classes

Class	mAP@.5	mAP@.5:.95
all	0.922	0.634
D40	0.831	0.431
D20	0.901	0.569
D50	0.888	0.583
D10	0.942	0.701
D44	0.962	0.793
D00	0.936	0.573
D43	0.993	0.791

According to Table II, the overall accuracy of the model is about 92% when used custom pre-trained weights for the road damage dataset. mAP at 0.5 IoU threshold and 0.95 IoU threshold is given to better understand the performance. This proves that, with the mere original architecture of YOLOv5, the model is not only fast, but the accuracy is also high without any optimization methods integrated into it. The results show that the YOLOv5 can be used to detect the road damage type effectively.

TABLE III: Precision and Recall value for specific damage class

Class	Precision	Recall
all	0.905	0.872
D40	0.864	0.686
D20	0.903	0.862
D50	0.887	0.826
D10	0.939	0.906
D44	0.904	0.935
D00	0.894	0.887
D43	0.945	0.999

Table III shows that the overall precision and recall of the given model is 90.5% and 87.2% respectively. D43 road damage type was detected with high recall and precision, and the feature is also clear due to that fact that D43 (Cross walk blur) takes up significant section in the image.

Different type of training loss shown in figure 5. Box Loss can be defined as how good the center of an object is found and how good the predicted bounding box covers an object. In other words, its a loss that measures how well the predicted bounding boxes fit to the ground truth object. Classification loss gives an idea of how accurately the YOLOv5 algorithm can predict the correct class. It is clear from the figure that as the number of epochs increases,the loss value becomes smaller.

From the above results,we can conclude that the road damage detection and classification model is quite high on

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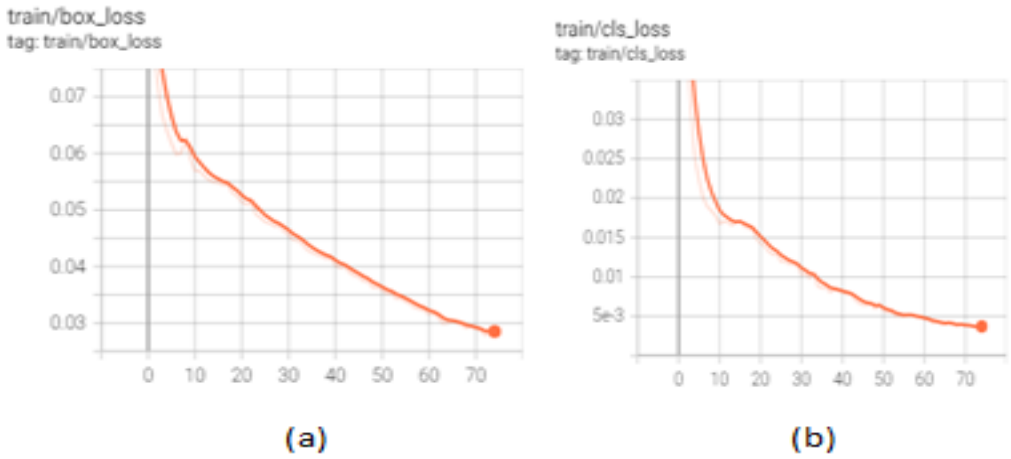


Fig. 5: Plots of the box loss and the classification loss

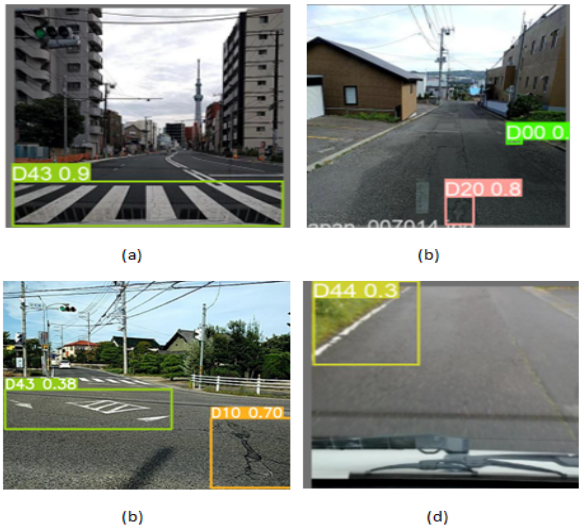


Fig. 6: Predicted Road Damage Images

performance. YOLOv5 also proven to be fast in real time and with above 90% accuracy, the proposed model can be deployed in real world applications. As part of future work, the dataset can include images taken from various countries. Figure 6 shows different road damage classes predicted by YOLOv5 model.

VI. CONCLUSION

A deep learning-based automated method for road damage detection and classification is nowadays timely needed for effective maintenance of road. To create an efficient and effective road damage detection model,the proposed system used YOLOv5 model to identify seven types of road damages. As YOLOv5 is the lightweight and latest version which uses Pytorch framework. It has a fast detection speed, thereby can effectively work in real-time detection. The YOLOv5 model has also the advantage of adaptive generation of anchor boxes that helps to accurately fix the bounding boxes on the road damage. model can effectively classify longitudinal crack and crosswalk blur. The Road Damage Detection model is built using YOLOv5 model which obtained an overall mAP of 92%.

The future work is to compare various one-stage detectors with the YOLOv5 model to better prove the performance of YOLOv5. This project can be used as a base model and the dataset can be expanded by including additional images from different countries under different lighting to better represent each damage category and improve the robustness of the overall detection system.

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